Aditya Suyal

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Education

Birla Institute of Applied Sciences Bhimtal, Uttarakhand
Bachelor of Technology in Computer Science *Aug 2023 – Aug 2027*

Technical Skills

Languages: Python, Java, C++, SQL, JavaScript, Rust

Model Architectures: Encoder-Only (BERT), Encoder-Decoder (T5, BART), Decoder-Only (GPT, LLaMA), Transformers, RNNs, LSTMs

AI/ML Frameworks: PyTorch, TensorFlow, LangChain, LlamaIndex, CrewAI, Autogen, LangGraph

LLM Technologies: GPT-4, Gemini Pro, Llama 2/3, Mistral, Stable Diffusion, LoRA, QLoRA

Vector Databases: FAISS, Pinecone, ChromaDB, MongoDB

Cloud & Deployment: AWS (EC2, Lambda), Docker, Kubernetes

AI Techniques: RAG, Multi-Agent Systems, Memory-Augmented LLMs, MLOps, LLMOps

Experience

AI Agent Development Intern Dec 2025 – Feb 2026
Shell Global Pvt. Ltd. *Remote*

– Built production-grade multi-agent AI systems using LangChain and CrewAI, reducing manual processing by 65%.
– Engineered autonomous tool-calling workflows using GPT-4 and Llama, achieving 88% task accuracy.

Generative AI Intern Nov 2025 – Dec 2025
Analytics Vidhya *Remote*

– Developed RAG pipelines processing 10K+ documents with 92% retrieval accuracy.
– Built scalable AI APIs serving 500+ concurrent users with optimized FAISS embeddings.

Projects

Hybrid Multi-Agent Conversational AI Architecture | *LLM Systems Research* | *Python, CrewAI, Llama 3.3 (Groq)* 2026

– First-author research paper (March 2026) introducing a 10-agent modular conversational framework addressing limitations of monolithic LLM inference including domain specialization trade-offs, absence of structured verification, and limited personalization.
– Combined supervisor-based routing, competitive multi-agent debate (LLM-as-a-Judge), and expertise-aware personalization; achieved **+11.9% accuracy improvement** over single-model baseline (150-query blind evaluation, $p < 0.001$).
– Improved beginner comprehension by **+21 percentage points** and reduced unnecessary tool API calls by 31.4% through context-aware decision agents.

Memory-Augmented LLM using LSTM (Hybrid Intelligence) | *Python, PyTorch, LSTM, Transformer-based LLMs* 2026

– Designed hybrid architecture combining Transformer-based LLM reasoning with LSTM-based temporal memory for long-term conversational context retention.
– Stored interaction embeddings from user sessions, processed through LSTM to generate compressed dynamic memory states, and injected memory summaries back into the LLM as contextual bias.
– Proposed separation of reasoning (LLM) and sequential memory modeling (RNN/LSTM), enabling scalable long-term adaptation without increasing Transformer context window size.

Private RAG Document QA System | *LangChain, FAISS, Gemma-2B, Streamlit* 2025

– Built secure RAG system with **Hybrid Retrieval** (dense + sparse BM25) and **Multi-Query RAG** enabling context-aware Q&A over private PDFs using sentence-transformers and Hugging Face API.
– Processed 5K+ queries with 90% relevance score; reduced query latency from 3s to 0.4s through optimized FAISS indexing and multi-query expansion for improved document recall.

Agentic AI Multi-Agent System for EV Research | *Python, Agno, Groq, AWS* 2025

– Architected autonomous multi-agent system (Supervisor -Co-workers) with 5+ specialized LLM agents for EV market and policy analysis, reducing manual research effort by 70%.
– Integrated search Tools and automated reporting pipelines deployed on AWS.

Achievements & Certifications

Awards: 1st Place TechFest AI in Education | 3rd Place Grafest AI in Healthcare | 2× State AI Hackathon Finalist

Certifications: Multi-Agent Systems (DeepLearning.AI) | Azure AI Fundamentals | HackerRank SQL Advanced | Applied AI Lab (WorldQuant University, USA) | Data Science Lab — 16-Week Program (WorldQuant University, USA)